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COP4493 - AWS Academy Cloud Computing Fundamentals

Project-1

URL of Website: lb-website-hosting-burakdmrc-1713133902.us-east-1.elb.amazonaws.com

AWS Account ID: 895084669974

GitHub Repository URL: https://github.com/burakdemirci03/cop4493_website

S3 Bucket Creation

1. **Bucket Type:** Select “General Purpose”.
2. **Bucket Name:** Any name to identify the bucket, I gave “s3-website-hosting-burakdmrc”.
3. Other configurations can stay same, then proceed with “Create Bucket” at the end of the page.

The screenshot shows the 'General configuration' tab for creating a new S3 bucket. The 'AWS Region' is set to 'US East (N. Virginia) us-east-1'. Under 'Bucket type', the 'General purpose' option is selected, which is recommended for most use cases. The 'Directory' option is also visible. The 'Bucket name' field contains 's3-website-hosting-burakdmrc'. Below this, there is a note about bucket name constraints and a 'Choose bucket' button. The format 's3://bucket/prefix' is shown at the bottom.

4. After creating the bucket, select the bucket and then upload the HTML, CSS and JS files using top left button.
5. Then go to “Properties” from top bar and find the Static Website Hosting in the bottom of the page. And click to “Edit” button.
6. **Static website hosting:** Enable the radio button.
7. **Hosting Type:** Select “Host a static website” area.
8. **Index Document:** Set the same HTML file as the name. Mine was “index.html”, I set it so. If an error endpoint is given, it can also be named the same way.
9. Don’t need to change anymore, so “Save Changes”.

The screenshot shows the 'Static website hosting' configuration page. The 'Static website hosting' section has the 'Enable' radio button selected. Under 'Hosting type', the 'Host a static website' option is selected. A note indicates that content must be publicly readable for website access. The 'Index document' field is set to 'index.html', and the 'Error document' field is set to 'error.html'.

VPC Creation

1. **Resources to Create:** Select “VPC and more”. So, while creating VPC, NAT and subnets are also created by AWS.
2. **Name Tag Auto-Generation:** Enable “Auto-generate” and name anyway you wish, I named myself “vpc-website-hosting-burakdmrc”.
3. **IPv4 CIDR Block:** I gave the address space as “10.128.0.0/16”. But any other suitable spaces can be given.
4. **IPv6 CIDR Block:** Select “No IPv6 CIDR Block”.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
vpc-website-hosting-burakdmrc

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.128.0.0/16 65,536 IPs
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)
Default

► Encryption settings - optional

5. **Number of Availability Zones (AZs):** Select **2**. And choose two different AZs, I choose **us-east-1a** and **us-east-1b**.
6. **Number of Public/Private Subnets:** Select **2** for both public and private subnets.
7. **Subnets CIDR Blocks:** Is optional to modify. I named;
 - a. **Public Subnet CIDR Block in us-east-1a:** 10.128.0.0/24
 - b. **Public Subnet CIDR Block in us-east-1b:** 10.128.2.0/24
 - c. **Private Subnet CIDR Block in us-east-1a:** 10.128.1.0/24
 - d. **Private Subnet CIDR Block in us-east-1b:** 10.128.3.0/24

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 | **2** | 3

▼ Customize AZs

First availability zone
use1-az6 (us-east-1a)

Second availability zone
use1-az1 (us-east-1b)

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 | **2**

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 | **2** | 4

▼ Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a
10.128.0.0/24 256 IPs

Public subnet CIDR block in us-east-1b
10.128.2.0/24 256 IPs

Private subnet CIDR block in us-east-1a
10.128.1.0/24 256 IPs

Private subnet CIDR block in us-east-1b
10.128.3.0/24 256 IPs

8. **NAT Gateways**: Select “Zonal” first and then “1 per AZ” in order.

NAT gateways (\$) - updated Info
NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.

None | Regional - new | **Zonal**

NAT gateways (\$) Info
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

In 1 AZ | **1 per AZ**

VPC endpoints Info
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None | **S3 Gateway**

DNS options Info
☒ Enable DNS hostnames
☒ Enable DNS resolution

9. Then proceed with “Create VPC” finally.

PS If I do remember correctly, the **Subnets, Route Tables, Internet Gateway and NAT Gateways** are also configured properly while creating the VPC.

If not configured automatically;

- **for Route Table for Public Subnet**: Any IPv4 address should target to the Internet Gateway (*0.0.0.0/0 -> igw-xxxxxx*).
- **for Route Tables for Private Subnets**: Any IPv4 address should target to its corresponding NAT Gateway (*0.0.0.0/0 -> nat-xxxxxx*).

EC2 Instance Creation

1. **Name**: Select any name you prefer, I named “ec2-website-hosting-burakdmrc”.
2. **Amazon Machine Image (AMI)**: Do not need to change the default “Amazon Linux” and “Amazon Linux 2023 kernel-6.1 AMI” configuration.

Name and tags Info
Name
ec2-website-hosting-burakdmrc Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian
aws Mac ubuntu Microsoft Red Hat SUSE Linux Debian

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

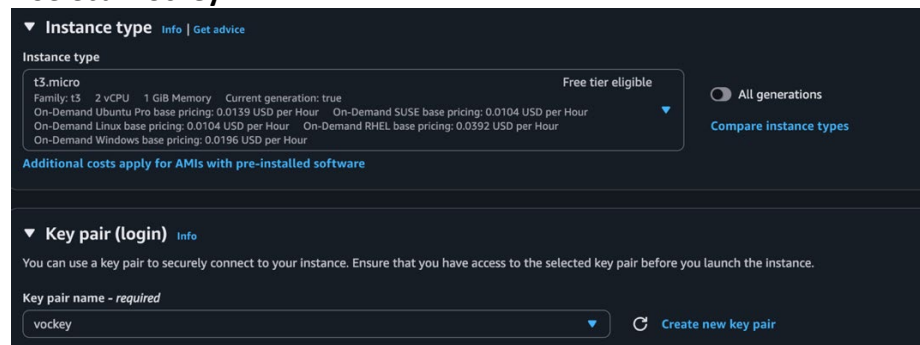
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.1 AMI
ami-0fa3fe0fa7920f68e (64-bit (x86), uefi-preferred) / ami-0d4a28e5df2d25176 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 9.20251117.1 x86_64 HVM kernel-6.1

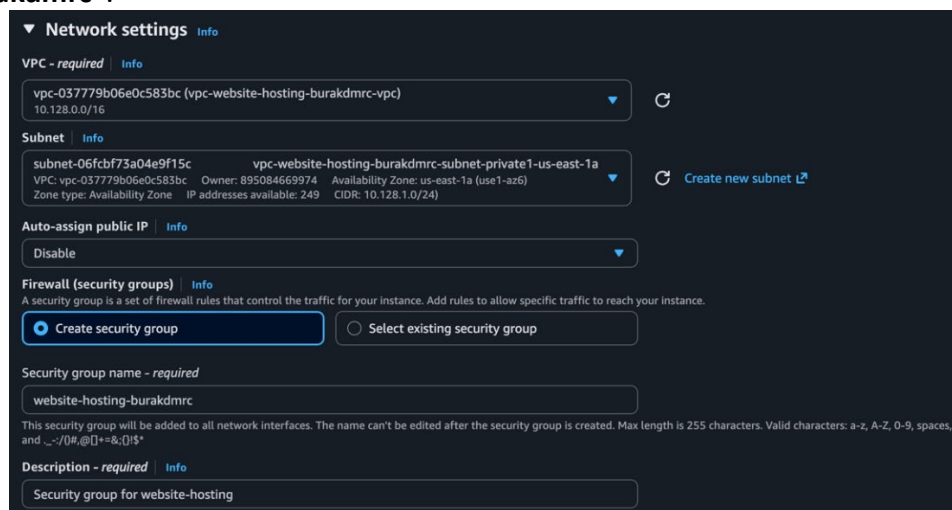
Architecture Boot mode AMI ID Publish Date Username
64-bit (x86) uefi-preferred ami-0fa3fe0fa7920f68e 2025-11-17 ec2-user Verified provider

3. **Instance Type:** The default **t3.micro** can also remain the same.
4. **Key Pair:** Select “**vockey**”.



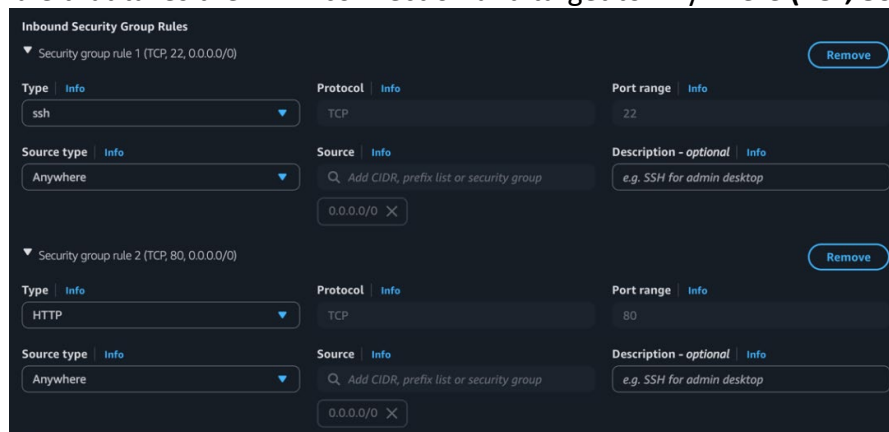
The screenshot shows the AWS Management Console configuration page for an EC2 instance. Under the 'Instance type' section, 't3.micro' is selected, which is marked as 'Free tier eligible'. Below this, pricing information for various operating systems is listed. In the 'Key pair (login)' section, a dropdown menu shows 'vockey' as the selected key pair. A 'Create new key pair' button is visible next to the dropdown.

5. **VPC:** Select the recently created VPC, “**vpc-website-hosting-burakdmrc-vpc**”.
6. **Subnet:** Choose any private subnet of given VPC, I choose us-east-1a private subnet “**vpc-website-hosting-burakdmrc-subnet-private1-us-east-1a**”.
7. **Auto-assign Public IP:** Disable the setting.
8. **Firewall (Security Group):** We do not have any security group yet so choose “**Create Security Group**”.
9. **Security Group Name:** Name any way you prefer, I named “**website-hosting-burakdmrc**”.



The screenshot displays the 'Network settings' section of the AWS Management Console. The 'VPC' dropdown is set to 'vpc-037779b06e0c583bc (vpc-website-hosting-burakdmrc-vpc)'. The 'Subnet' dropdown is set to 'subnet-06fcbf73a04e9f15c vpc-website-hosting-burakdmrc-subnet-private1-us-east-1a'. The 'Auto-assign public IP' setting is set to 'Disable'. Under the 'Firewall (security groups)' section, the 'Create security group' radio button is selected. The 'Security group name' field contains 'website-hosting-burakdmrc', and the 'Description' field contains 'Security group for website-hosting'.

10. **Inbound Security Group Rules:** Keep the default ssh rule (**TCP, 22, 0.0.0.0/0**) and add a new rule that takes the HTTP connection and target to Anywhere (**TCP, 80, 0.0.0.0/0**)



The screenshot shows the 'Inbound Security Group Rules' configuration page. It lists two rules. Rule 1 is for 'ssh' with protocol 'TCP', port range '22', and source 'Anywhere' (0.0.0.0/0). Rule 2 is for 'HTTP' with protocol 'TCP', port range '80', and source 'Anywhere' (0.0.0.0/0). Each rule has a 'Remove' button in the top right corner.

11. Then launch the EC2 instance.

12. We created this instance on only one subnet so, when created the first EC2 instance, select the created instance (given “**ec2-website-hosting-burakdmrc**”) and in top right bar select “Actions” and “Images and Templates”, then click “**Launch more like this**”. Then in opened tab;

- **Subnet:** Change the subnet to the other private subnet. I choose private1-us-east-1a in first instance, so in this instance I changed to “**vpc-website-hosting-burakdmrc-subnet-private2-us-east-1b**”.

(I did not launch another instance, which would cost too much in long-term)

13. In left scrollable are click “**Target Groups**”, then in top right click “**Create Target Group**”. We will need a target group in Load Balancer.

14. **Target Type:** Keep the setting in **Instances**.

15. **Target Group Name:** Give a unique name for Target Group, I gave “**tg-website-hosting-burakdmrc**”.

16. **Protocol & Port:** Choose **HTTP** protocol and port **80**.

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.
tg-website-hosting-burakdmrc
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 28/32

Protocol
Protocol for communication between the load balancer and targets.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

17. **IP Address Type:** Keep in **IPv4**.

18. **VPC:** Select the previous VPC, “**vpc-website-hosting-burakdmrc-vpc**”.

19. **Protocol Version:** Keep on **HTTP1**.

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-037779b06e0c583bc (vpc-website-hosting-burakdmrc-vpc)
10.128.0.0/16

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

20. Then proceed to “Next” page.

21. **Register Targets:** Select the recently created EC2 instance, which was “**ec2-website-hosting-burakdmrc**” for me. Then click “Include as pending below” button and proceed to “Next” page where you would end up in “Create Target Group”.

22. In left scrollable are click **“Load Balancer”**, then in top right click **“Create Load Balancer”**.
23. Select **Create** from **Application Load Balancer** section in **Load Balancer Types**.
24. **Load Balancer Name**: Give a unique name for Load Balancer, I gave **“lb-website-hosting-burakdmrc”**.
25. **Scheme**: Select **Internet-facing**.
26. **Load Balancer IP Address Type**: Choose **IPv4**.

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

lb-website-hosting-burakdmrc

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme Info
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

27. **VPC**: Select the previous VPC, **“vpc-website-hosting-burakdmrc-vpc”**.
28. **Availability Zones and Subnet**: Select both subnets and its corresponding public subnets **“vpc-website-hosting-burakdmrc-subnet-public[x]-us-east-1[x]”**.

Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-037779b06e0c583bc (vpc-website-hosting-burakdmrc-vpc)

IP pools Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az5)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-06a464c25b945d23e
IPv4 subnet CIDR: 10.128.0.0/24

vpc-website-hosting-burakdmrc-subnet-public1-us-east-1a

☒ **us-east-1b (use1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-08646336a641a98b1
IPv4 subnet CIDR: 10.128.2.0/24

vpc-website-hosting-burakdmrc-subnet-public2-us-east-1b

29. **Security Groups**: Select the security group, which was created while creating VPC, for me it was **“website-hosting-burakdmrc”**.

Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

website-hosting-burakdmrc
sg-0ab38e839f9b3519 VPC: vpc-037779b06e0c583bc

30. **Protocol & Port**: Choose **HTTP** protocol and port **80**.
31. **Routing Action**: Select **“Forward to target groups”**.
32. **Target Group**: Select the recently created target group, **“tg-website-hosting-burakdmrc”**.

(Since I have used target group before, I cannot choose it while showing in another instance)

▼ Listener HTTP:80 Remove

Protocol: HTTP Port: 80
1-65535

Default action Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group Info
Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
Select a target group	1	100%
	0-999	

+ Add target group
You can add up to 4 more target groups.

Target group stickiness Info
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag
You can add up to 50 more tags.

33. Then continue with “Create Load Balancer” which is located the end of page.

Web Server Configuration

1. Select the EC2 instance and click to the “Connect” button at the top-right corner.
2. In instance connection try one of the following;
 - a. Go back to the instances page and select the instance, then proceed with Actions > Networking > Manage IP Address. Open up the collapsable Network Interface section (named like, “eth0: eni-[xxxx] - 10.128.1.0/24”) and turn on the “Auto-assign Public IP” option and Save. Then go back to Connection tab, select “**EC2 Instance Connect**” and “Connect using a Public IP”.
(turn off the “Auto-assign Public IP” when all configuration are done)
 - b. Connect with “**SSH Client**” using the instruction in readme manual on AWS Academy website.
 - c. If possible, connect using “**Session Manager**”.
(while doing project, most of time this option was not open for me but now it is open, even if I do not know why)
3. Create the Apache Server using following instructions;


```
sudo yum update
sudo yum install httpd
sudo systemctl start httpd
sudo systemctl enable httpd
```
4. Then load the objects -HTML, CSS and JS files- from S3 to the Apache server using;


```
sudo aws s3 sync s3://s3-website-hosting-burakdmrc /var/www/html
```

Website should be accessible by the DNS record of the load balancer now!